



SIM32ELA and SIM33ELA_Compatible_Desi gn

GNSS Module

SIMCom Wireless Solutions Limited

Building B, SIM Technology Building, No.633, Jinzhong Road
Changning District, Shanghai P.R.China

Tel: 86-21-31575100

support@simcom.com

www.simcom.com

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SIMCom Wireless Solutions Limited

Building B, SIM Technology Building, No.633 Jinzhong Road, Changning District, Shanghai P.R.China
Tel: +86 21 31575100
Email: simcom@simcom.com

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1. Introduction

This document is targeted for customers to understand the differences between SIM32ELA and SIM33ELA. Customers can use SIM32ELA or SIM33ELA module to design and develop applications quickly.

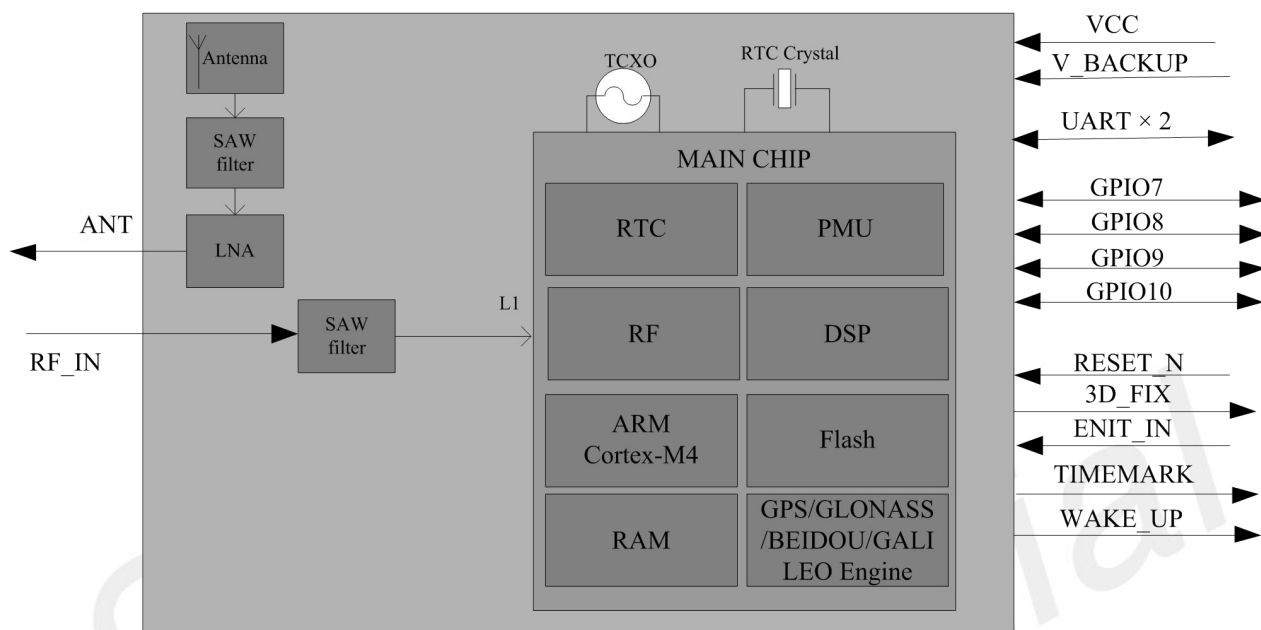


Figure 1: SIM32ELA functional diagram

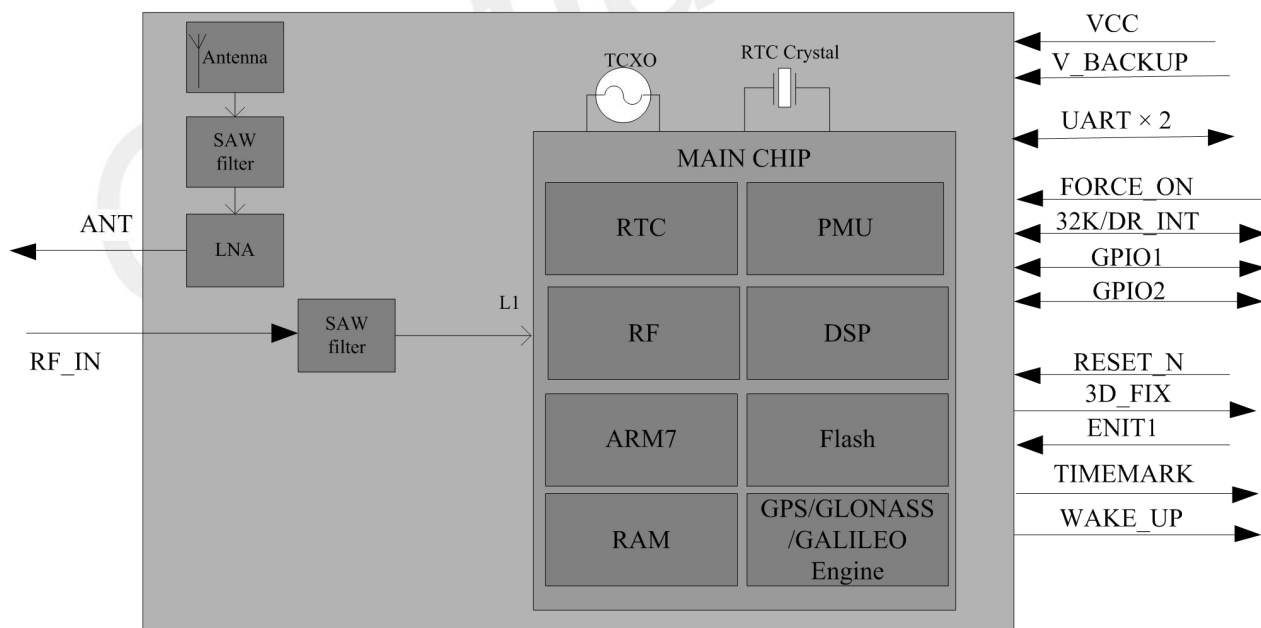


Figure 2: SIM33ELA functional diagram

2. PIN assignment

2.1 Pin Assignment Overview

The following figure shows the pin assignment of SIM32ELA and SIM33ELA.

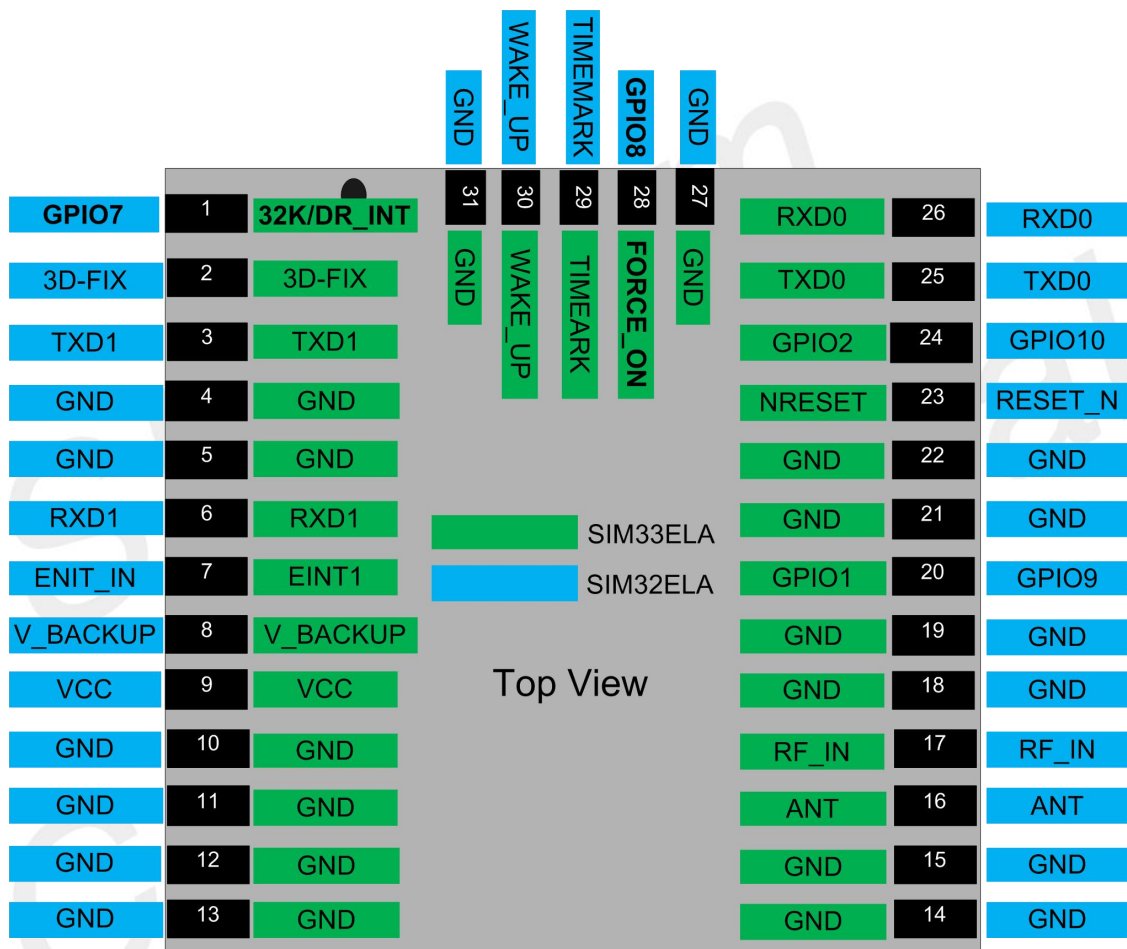


Figure 3: SIM32ELA and SIM33ELA pin assignment (Top view)

2.2 Differences Overview

Table 1: The differences overview

Functions	SIM32ELA	SIM33ELA
UART0	Support NMEA output and PMTK command input and update	Support NMEA output and PMTK command input and update

UART1	Support RTCM	Support RTCM
PPS	Support	Support
RESET	Support	Support
Sleep mode	Support	Support
RTC mode	Support	Support
Internal Antenna	Support	Support
Antenna detect	Not support	Not support
IO power domain	2.8V	2.8V
GPS satellite	Support	Support
BEIDOU satellite	Support	Not support
GLONASS satellite	Support	Support
GALILEO satellite	Support	Support

2.3 Differences of GNSS Performance

Table 2: The differences of GNSS Performance

Parameter	Description	SIM32ELA	SIM33ELA	Unit
		Type	Type	
Horizontal Position Accuracy¹	Automatic position	1	2.5	m
Dynamic Performance	Maximum Altitude	18000	18000	m
	Maximum Velocity	500	515	m/s
	Maximum Acceleration	4	4	G
Time To First Fix²	Hot start	0.46	1	s
	Warm start	15.40	26	s
	Cold start	15.46	28	s
A-GPS TTFF(EPO in flash mode)	Hot start	0.61	1.04	s
	Warm start	20.45	21.54	s
	Cold start	20.77	22.03	s
Sensitivity³	Autonomous acquisition(cold start)	-148.5	-147	dBm
	Re-acquisition	-158.5	-160	dBm
	Tracking	-167	-165	dBm
Power consumption⁴	Acquisition	13	25	mA
	Continuous tracking	13	20	mA
	Sleep current	200	320	uA
	RTC current	15	14	uA

NOTE

1. 50% 24hr static, -130dBm;
2. SIM33ELA with GPS(L1)+GLONASS mode, SIM32ELA with GPS(L1)+GLONASS+BEIDOU mode@-130dBm;
3. SIM33ELA with GPS(L1)+GLONASS mode, SIM32ELA with GPS(L1)+GLONASS+BEIDOU mode@-130dBm;
4. SIM33ELA with GPS(L1)+GLONASS mode, SIM32ELA with GPS(L1)+GLONASS+BEIDOU mode.

2.4 Differences of Electronic Characteristic

Table 3: The differences of electronic characteristic

Pin #	SIM32ELA		SIM33ELA	
	PIN name	Voltage range	PIN name	Voltage range
9	VCC	2.8~4.3V	VCC	2.8~4.3V
8	V_BACKUP	1.7~3.6V	V_BACKUP	2.0~4.3V
1	GPIO7	2.8V	32K/DR_INT	1.1V
28	GPIO8	2.8V	FORCE_ON	1.1V
30	WAKE_UP	1.8V	WAKE_UP	2.8V

NOTE

1. For details information, please refer to each HD guide

2.5 Differences of Pin Definitions

Table 4: Difference in Pin Definitions

Pin #	SIM32ELA	Description	SIM33ELA	Description
1	GPIO7	General purpose input/output, Reserved	32K/DR_INT	Reserved for wake-up interrupt(DR_INT default) Optionally 32768 Hz RTC clock output
7	EINT_IN	Wake up SIM32ELA from uart sleep, low active , default pull high	EINT1	This interrupt source could control the module come into or wake up from sleep mode. Low=come into sleep mode.High=wake up from sleep mode.

20	GPIO9	General purpose input/output, Reserved	GPIO1	General purpose input/output, Reserved
23	RESET_N	Reset the module, active low	NRESET	Reset input, active low, default pull-up
24	GPIO10	General purpose input/output, Reserved	GPIO2	General purpose input/output, Reserved
28	GPIO8	General purpose input/output, Reserved	FORCE_ON	Power control input used to force wake up from low power modes.

2.6 Differences of NMEA Command Definitions

Table 5: Difference of NMEA Command Definitions

SN	SIM32ELA	SIM33ELA
[1]	SIM32 Series_NMEA Message_User Guide_V1.00	MT3333 Platform NMEA Message Specification_V1.00

3. Recommended Footprint

3.1 Top and Bottom View

The following figures show top and bottom view of SIM32ELA and SIM33ELA.

There has no difference for footprint.



Figure 4: SIM32ELA and SIM33ELA top and bottom view

3.2 Recommended Stencil Design

SIM32ELA and SIM33ELA have the same recommended stencil design.

The recommended stencil design for SIM32ELA and SIM33ELA is shown as below.

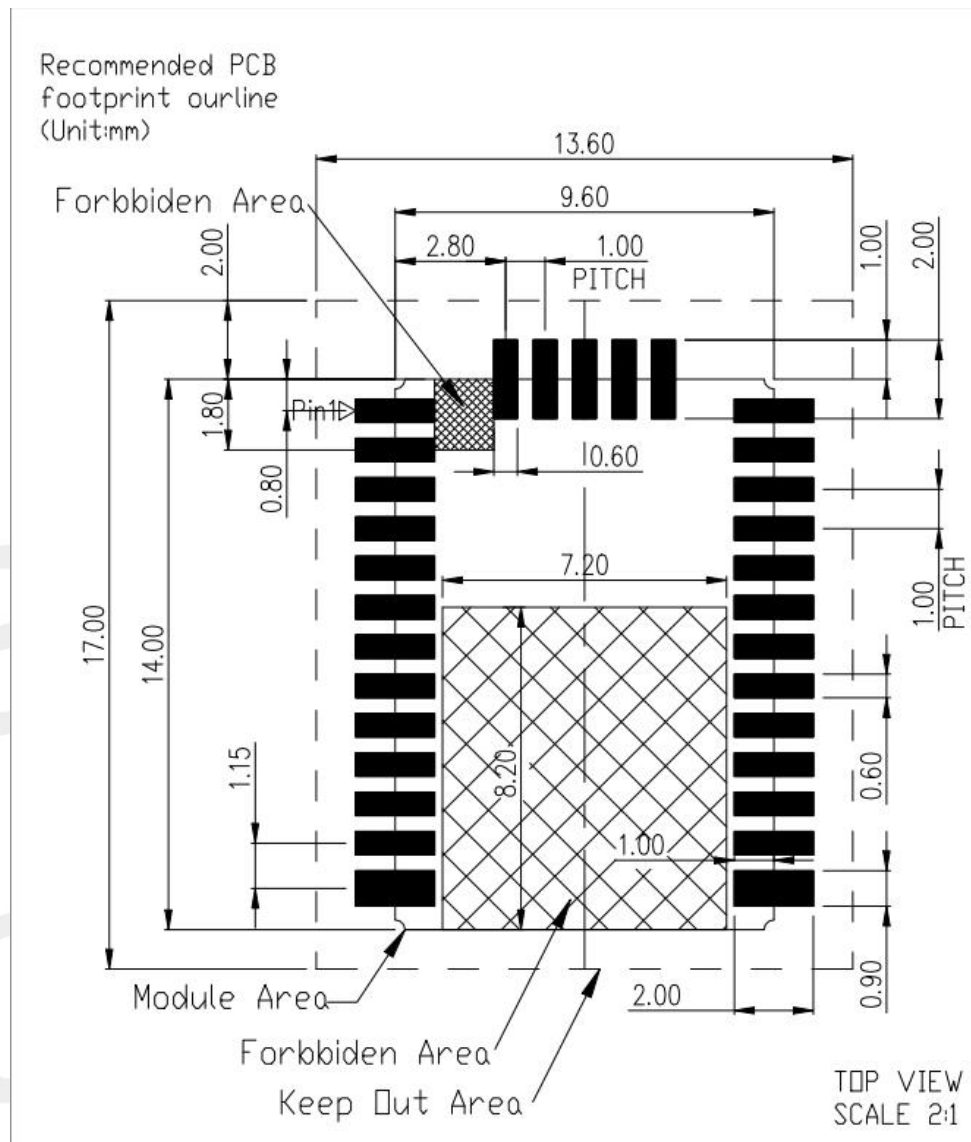


Figure 5: Recommended Stencil Design for SIM32ELA and SIM33ELA (Unit: mm)

4. Antenna Interface

SIM33ELA and SIM32ELA have an embedded GNSS Antenna and the signal is further amplified by internal Low Noise Amplifier (LNA), which is available at ANT output. The antenna signal ANT shall be connected externally to RF_IN Antenna Input signal via a short trace between pad ANT and pad RF_IN.

If using the embedded antenna, make sure the pad ANT be connected externally to RF_IN pad via a short trace with 50Ω impedance. The RF input signal path contains a first SAW band-pass filter before LNA, which provides excellent protection against out-of-band GNSS blocking caused by possible near-by wireless transmitters. Figure 6 shows the reference design circuit of SIM32ELA embedded antenna .

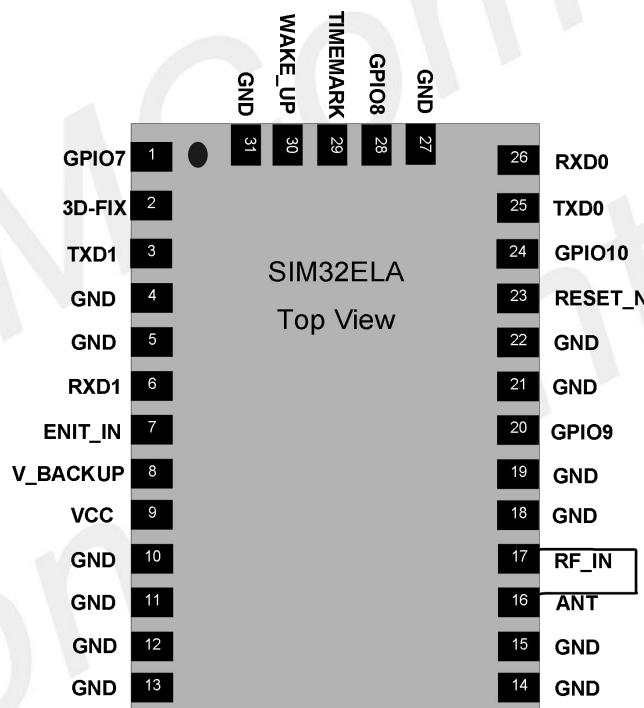


Figure 6: Reference design circuit for antenna embedded

5. Appendix

5.1 Related Documents

Table 6: Related documents

SN	Document name	Remark
[1]	SIM32LEA_Hardware_Design_V1.00	SIM32ELA Hardware Design Document
[2]	SIM33ELA_Hardware_Design_V1.02	SIM33LEA Hardware Design Document

5.2 Terms and Abbreviation

Table 7: Terms and Abbreviations

Abbreviation	Description
GNSS	Global Navigation Satellite System
NMEA	National Marine Electronics Association
IO	Input/Output
RF	Radio Frequency
RTC	Real Time Clock
RTCM	Radio Technical Commission for Maritime Services
UART	Universal Asynchronous Receiver & Transmitter
NC	Not connect
TTFF	Time To First Fix